

CMDistill: Cross-Modal Distillation Framework for AAV Image Object Detection

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Abstract—With the increasing intelligence of autonomous aerial vehicles carrying diverse mission payloads, the target detection domain consists mainly of single-modal and multimodal approaches with diverse and complex combinations. It is challenging to achieve an optimal tradeoff between expensive and complex models of multimodal detectors, which are difficult to deploy directly on autonomous aerial vehicles (AAVs), and the limited detection accuracy of single-modal detectors. To overcome this limitation, we developed a cross-modal target detector called CMDistill. Specifically, we designed an effective distillation loss method based on three components. First, to reduce differences in feature knowledge across modal interlayers, we designed a Pearson correlation coefficient to constrain negative knowledge. Second, we modeled the relational cues between features by computing the affinity matrices of the deeper semantic features of the teacher–student model to convey relational knowledge more accurately. Finally, the target bounding boxes and classification information predicted by the output of the teacher were passed to the student. Experimental results on the aerial vehicle detection dataset revealed that CMDistill achieved an optimal performance with an average accuracy of 74% on the RGB-only target detection task using fewer computational resources to achieve a detection performance comparable with that of multimodal approaches.

Index Terms—Cross-modality, knowledge distillation (KD), object detection, autonomous aerial vehicle (AAV).

I. INTRODUCTION

IN RECENT years, owing to high mobility, small size, and low cost, autonomous aerial vehicles (AAVs) have been widely used in fields, such as battlefield reconnaissance, damage assessment, smart agriculture, electric power inspection, and geological disaster rescue [1], [2]. In addition, advances in artificial intelligence technology significantly improved the environmental perception and autonomous decision-making capabilities of AAVs. The development of computer vision technology has significantly promoted the transformation of AAV environment perception tasks, thus attracting widespread attention from both academia and industry [3]. With the development of sensor technologies, the tasks performed by AAVs are more diversified, and recent research underwent a gradual shift from single-modal to

multimodal target detector designs, with the objective to obtain complementary feature information from multiple sensors and improve the generalization of models in different application scenarios. RGB sensor technology is relatively advanced and images conform to the sensory information processed by the human eye with clear texture details. Moreover, RGB target detectors can classify and localize targets accurately in low-noise scenes [4].

Although RGB images have clear textural details and are readily interpretable and simple to understand, they are susceptible to adverse weather conditions (e.g., dense fog and lighting conditions). In these challenging scenarios, it is insufficient to rely on single-modal images to achieve target detection, and the addition of auxiliary information from different modalities [e.g., thermal imaging information from infrared (IR) sensors] is necessary. Therefore, to improve the robustness of AAV target detection in complex scenarios, such as nighttime and foggy environments, multimodal detection frameworks have gradually emerged as a major research topic. Such frameworks aim to exploit the robust and information-rich features of multisource sensors to improve detection performance [5], [6], [7]. Gao et al. [8] developed a spatial–spectral perception network to extract dominant and cross-cutting information from different data sources. Zhao et al. [9] investigated the learning of more discriminative spatiotemporal distribution features from multitemporal remote sensing data. Wang et al. [10] individually extracted the features of the hyperspectral and complementary modes [X modes, including multispectral, synthetic aperture radar, and light detection and ranging (LiDAR)]. Notably, the information between the complementary modes was fully fused and enhanced. Although multimodal approaches exhibit high performances, there are several issues associated with their application to AAV edge-computing platforms, as follows.

- 1) *High device cost*: Given that IR sensors require a focal plane array and unique glass lens, their cost is significantly higher than that of RGB sensors, and it is impractical for all AAVs to carry RGB-thermal IR (RGB-T) sensors.
- 2) *Additional network design and computational overhead*: As shown in Fig. 1, a multimodal target detector requires the alignment of different sensor data, which involves millions of computations (more than 52.3 GFLOPs). However, the energy and computational capabilities of AAVs are limited, and it is difficult for such detection frameworks to be widely deployed in AAVs for real-time target detection.

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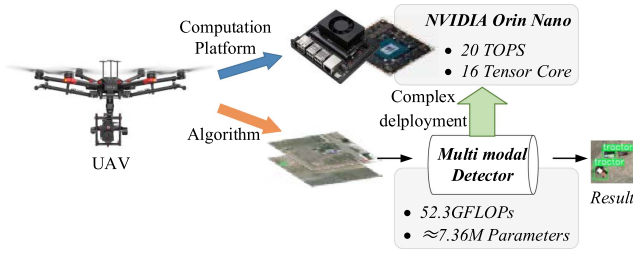


Fig. 1. Key issues associated with deploying multimodal detection frameworks on AAVs: The artificial intelligence performance of the SOTA edge-computing platform NVIDIA Orin Nano is 20 trillion operations per second and the computational complexity of most multimodal detectors is more than 52.3 GFLOPs, making it difficult to meet the real-time detection requirements of AAVs.

Therefore, for AAV platforms with limited computational resources, the design of efficient compressed models is necessary to maintain high detection accuracy with fast inference performance. Pruning, quantization, and knowledge distillation (KD) [11] are commonly used model compression techniques. KD is a teacher–student training paradigm that migrates key knowledge from a complex teacher model to a simple student model to achieve a tradeoff between inference speed and detection accuracy without changing the overall model design [12]. As an emerging general model compression architecture, KD is currently a popular research topic for classification and target detection applications. Accordingly, the form of knowledge transfer in KD has transitioned from an initial logical knowledge transfer to intermediate feature knowledge, relational feature knowledge, and structural feature knowledge transfer [13], [14]. However, most KD approaches investigate the same data source inputs. In particular, the inputs for the teacher and student models are of the same modality, and the student model can learn the knowledge representation of the teacher model effectively. For the cross-modal KD (CKD) of teacher and student models, the differences between modalities are detrimental to the transfer of critical knowledge, and it is probable for the detection accuracy of the student model to be negatively affected.

The performance bottleneck observed in CKD can be attributed to three main aspects. The first is the difficulty of transferring cross-modal knowledge. Existing approaches [15], [16], [17] transfer intermediate feature knowledge without considering the variability of intermediate-layer features across modalities for the teacher and student. The unprocessed intermediate-layer knowledge is transferred to the student model, which may not be valuable or may even produce misleading information. Second, constant relational mapping between the layers of the teacher and student models is key for the student models to learn the accurate relational knowledge from the teacher models. Capturing the Jacobi matrix of network-layer mapping similarities and using radial basis functions to compute interlayer correlations were demonstrated to improve student model performance through relational characterization, and a failure to accurately characterize cross-modal relational knowledge transfer may have a negative impact [18]. Finally, the approach of considering only the tag knowledge predicted by the output layer of the teacher model for delivery [19], [20], although simple, provides

limited information with relative uncertainty for the CKD task and should be combined with other distillation techniques to improve the detection performance of CKD student models.

Based on the analysis above, this article proposes a model called CMDistill to support CKD target detection and address the potential challenges of CKD detection in AAV images. The proposed CKD target detection framework includes three components for transferring feature and relationship knowledge in intermediate layers, as well as key knowledge for prediction. First, to reduce the potential noise effects of differences in feature knowledge in the intermediate layers of different modalities, this article proposes a Pearson correlation coefficient (PCC) to constrain negative knowledge and relax the constraints on the feature map size during distillation. Second, affinity matrices of the high-level semantic features of the teacher–student models are computed to model the relational cues between features to convey the relational knowledge of the teacher–student models more accurately. Finally, the target bounding boxes and classification information predicted by the output layer of the teacher model are transferred to the student model. It should be noted that the objective of CMDistill is to improve the distillation of the CKD student model by combining the knowledge of the features in the middle layer, relationships, and logic in the output layer to better transfer the knowledge of the teacher model to the student model.

The main contributions of this study can be summarized as follows.

- 1) We propose a CKD target detection paradigm called CMDistill, which is a lightweight and highly accurate model that can be deployed on a AAV to achieve the accurate detection accuracy and efficient inference speed.
- 2) We designed a PCC for feature distillation to transfer cross-modal feature knowledge. Semantic-level relational distillation (SLRD) was used to establish relational cues between the different feature modalities of the teacher–student models to transfer relational knowledge. The objective of intersection over union (IoU) based binary classification logic distillation (IBCLD) is to transfer bounding box regression knowledge and soft-target knowledge from the teacher model to the student model. The combination of three different levels of distillation improves the performance of single-modal detectors by transferring knowledge from different modalities and incurs no additional cost to the inference process.
- 3) We compared CMDistill with other state-of-the-art (SOTA) methods on the vehicle detection in aerial imagery (VEDAI) dataset, and the experimental results demonstrated that CMDistill achieved a superior detection performance. We conducted an extensive ablation study to demonstrate the effectiveness of each component of the proposed CMDistill model.

The rest of this article is organized as follows. Section II presents a review of the related studies on target detection and KD for AAV images. Section III presents the details of the proposed method. Section IV presents the experimental details and an analysis of experimental results. Finally, Section V concludes this article.

II. RELATED WORKS

A. AAV Target Detection

With the rapid development of computer vision technology, the use of convolutional neural networks to localize and classify AAV image targets has become a popular research topic. Existing target detection methods are typically classified into three categories, namely two-stage, one-stage, and end-to-end transformer methods. Two-stage methods (e.g., region CNN [22] (R-CNN) and faster R-CNN [23]) first generate regions of interest and then use subregion to fine-tune target bounding boxes and classification outputs. Numerous studies demonstrated the suitability of this approach for applications in scenarios where algorithmic real-time requirements are low and detection accuracy requirements are high. One-stage methods (e.g., single-shot multibox detector [24] and you-only-look-once (YOLO) [25]) handle the detection task as a regression problem by directly generating the classes and bounding box positions of targets. This approach maintains a high detection accuracy with a high inference speed and is suitable for use on AAV edge-computing platforms. End-to-end transformer methods (e.g., detection transformer (DETR) [26] and conditional-DETR [27]) handle the detection problem as an ensemble prediction problem and obtain target results directly without postprocessing. Although this approach improves model efficiency, the associated architectures have complex parameters and are not readily deployed on AAVs.

The above-mentioned generalized target detection models demonstrated high performances on the COCO [28] dataset. However, target detection in AAV aerial images is subject to scale variations, a significant number of small targets, complex image backgrounds, and significant variations in lighting conditions, thus further increasing the difficulty of target detection. Xu et al. [29] proposed a novel adaptive zoom network with flexible selective amplifiers that can be extended for target detection tasks. Zhu et al. [30] focused on dense targets in images using a convolutional block attention model and designed transformers as prediction heads to detect objects at different scales. Narayanan et al. [31] constructed the first dataset of AAV real fog scenes to advance the development of AAV image defogging methods and target detection algorithms under dense fog occlusion conditions. Ge et al. [32] proposed a two-stage framework using coarse foreground detection of dense target regions and fine detection of contextual relationships. Li et al. [33] proposed a two-stage progressive refinement process to detect occluded targets in AAV images based on local aggregation properties. Zhang et al. [34] proposed a method based on self-attention control and feature fusion to overcome the limitations associated with diverse backgrounds and small target sizes in AAV images. Zhang et al. [35] proposed a coarse-to-fine evolutionary agent detection framework that optimally scales to detect targets in complex images more effectively. Visible images are susceptible to lighting conditions, and several researchers considered target detection approaches using both RGB and IR modalities to improve the performance of target detection algorithms by leveraging the feature complementarity between different modalities. Sun et al. [36] constructed a AAV RGB-IR dataset spanning

a rich set of scenarios, which is referred to as DroneVehicle, containing 28439 image pairs. In addition, they proposed a framework to improve the performance of cross-modal vehicle detection in complex environments. Song et al. [37] constructed an unaligned RGB-T dataset called AAV RGB-T 2400, which contains many challenging scenes, and proposed a novel target search scheme.

Considering the above, these analyses demonstrate that most single-modal detectors demonstrate high performances with respect to high-definition images. However, in several challenging scenarios, such as low-light conditions, the detection performance of single-modal methods is unsatisfactory and potential application scenarios are limited, as shown in Fig. 2(a). Using the characteristics of different modalities to improve the target detection performance of single-modal models is a viable solution. It should be noted that RGB-T multimodal fusion can leverage complementary information. However, the high complexity of such models increases the overhead and memory requirements of AAV edge-computing platforms, which exacerbates the difficulty of edge-computing deployment, as shown in Fig. 2(b). Therefore, the proposed model can take the advantage of the complementary information of different modalities without increasing complexity to support single-modal detectors, as shown in Fig. 2(c). Considering the energy consumption and arithmetic constraints of AAV edge-computing platforms and the tradeoff between the model complexity and detection performance, we considered the design of efficient AAV image target detection from two perspectives. First, it is expected that the designed network can increase the precision of single-modal detectors by exploiting the complementary features of different modalities without increasing the complexity of the network structure to reduce the burden of AAV edge computing. Therefore, we adopted CKD to improve the robustness and accuracy of the proposed model. Second, it is expected that CKD can transfer knowledge between teacher and student models in different modalities, and that this approach can serve as a novel paradigm for single-modal target detection in AAVs, thus improving the generalization of single-modal target detection to different application scenarios.

B. Knowledge Distillation

KD was originally used in compression modeling [13], where the teacher model is larger and transfers knowledge to the student model. Most methods in this area were developed for image classification tasks [13], [38] and were seldom used for target detection tasks. Chen et al. [19] first applied cue learning and KD to a multiclass target detection task. Subsequently, an increasing number of studies used KD to improve target detection performance [19], [39], [40]. For target detection tasks, KD is typically classified into three categories of feature-, relationship-, and logic-based KD [12]. Feature-based KD expresses transfer knowledge using the high-dimensional feature layer of the teacher model, which is feature rich. However, it is challenging to integrate knowledge from feature layers to achieve complementary effects [14]. Relationship-based KD accounts for the differences between feature relationships and is focused on providing a constant relationship mapping to enable

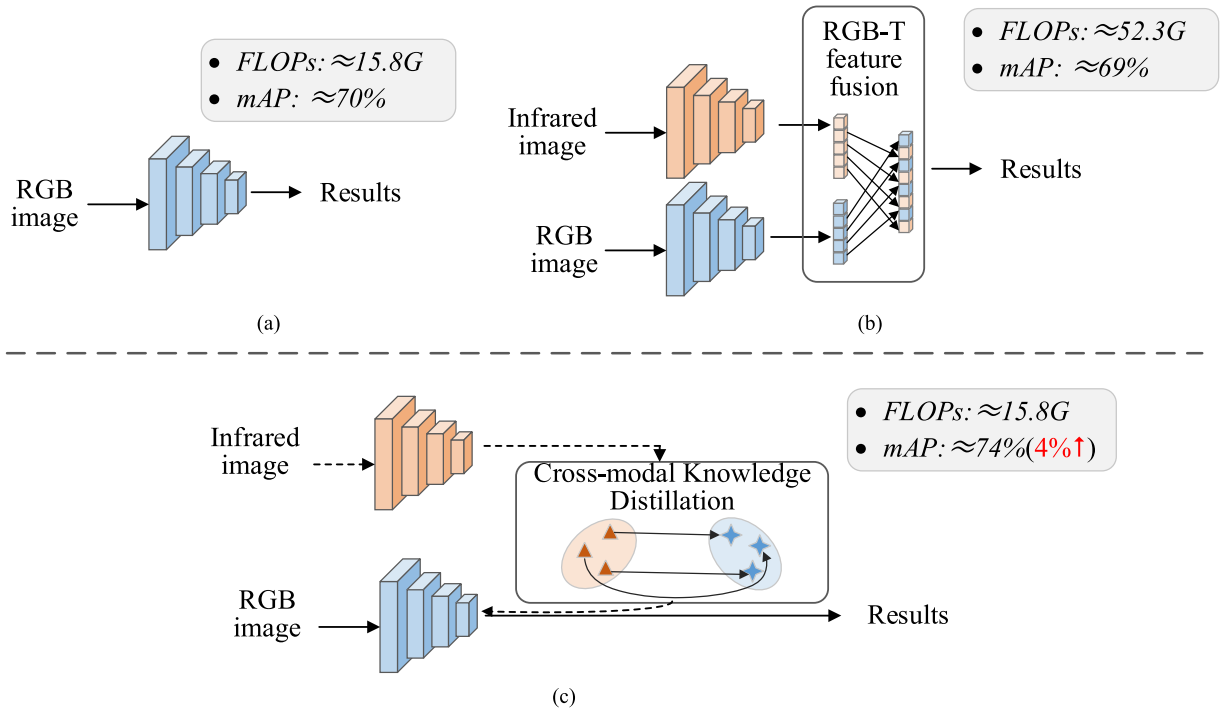


Fig. 2. Comparison of different target detectors. (a) Early single-modal detectors using an RGB image as an input source. (b) Existing RGB-T multimodal detectors requiring both RGB and IR images for training and inference [21]. (c) Proposed CKD target detector employing both RGB and IR images for training and only RGB single-modal images for inference. (a) RGB-based method. (b) RGB-T-based method. (c) Ours.

the student to accurately learn the relationship knowledge of the teacher. However, it is limited in which it is relatively difficult to model certain types of relationships [41]. Logic-based KD is a simple and effective KD approach for improving the effectiveness of training a student model using teacher predictions as soft labels. In particular, the limitation is that the labeled predictions outputted by the model express limited information [20].

The data are generally available in multiple modalities, and the data in various modalities can enrich overall knowledge. Common feature information among modalities can help strengthen the essential representations of data, and dissimilar feature information can help differentiate the feature performances of data. Therefore, CKD was proposed to help student models learn different modalities of knowledge conveyed by teacher models such that a single-modal model can achieve multimodal performance. This has emerged as a hot research topic across a wide range of fields [42], [43]. Albanie et al. [42] first proposed the most representative approach to emotion recognition in CKD. Wang et al. [43] designed EvDistill to fully exploit the potential correlations in paired modalities. Undistilled [44] is a generalized CKD framework to increase the accuracy of single-modal 3-D object detection. Notably, DistillBEV [45] utilizes effective balancing strategies to force students to focus on learning the most important features from teachers. Li et al. [46] proposed a CKD target detector with selective features and adaptive predictive imitation to convey valuable predictive knowledge.

However, most CKD methods do not fully consider the differences between modalities. Considering the CKD target detection

task, the teacher and student models are different modal inputs, namely, IR and RGB, respectively. Visible images contain clear texture details and IR images capture the thermal information of targets. These different modalities exhibit different features, and transferring the knowledge of modal features with large differences between teacher and student models is key to improving accuracy. To solve this problem and improve the recognition performance of the CKD student model, we considered the design of CMDistill from three perspectives. First, we aimed to perform knowledge transfer in a manner that makes use of multiscale feature maps with rich spatial information to the greatest extent possible. Therefore, PCCs were designed to transfer the knowledge of intermediate-layer features. Second, we modeled the relational cues of midlevel features and computed the affinity matrices of high-level semantic features to transfer key relational knowledge. Finally, localization and categorization information were transferred into simple and effective prediction boxes in the output layer of the teacher. We aimed to combine the feature, relational, and logical knowledge transferred from feature and output layers to reduce the noise effects generated by different modal distillations, transfer more critical and effective knowledge to the student, and improve the distillation effect.

III. METHODOLOGY

Based on the comprehensive analysis of the difficulties and challenges associated with cross-modal target detection for AAVs, we designed CMDistill with three main components, namely, a feature layer, relationship layer, and response layer,

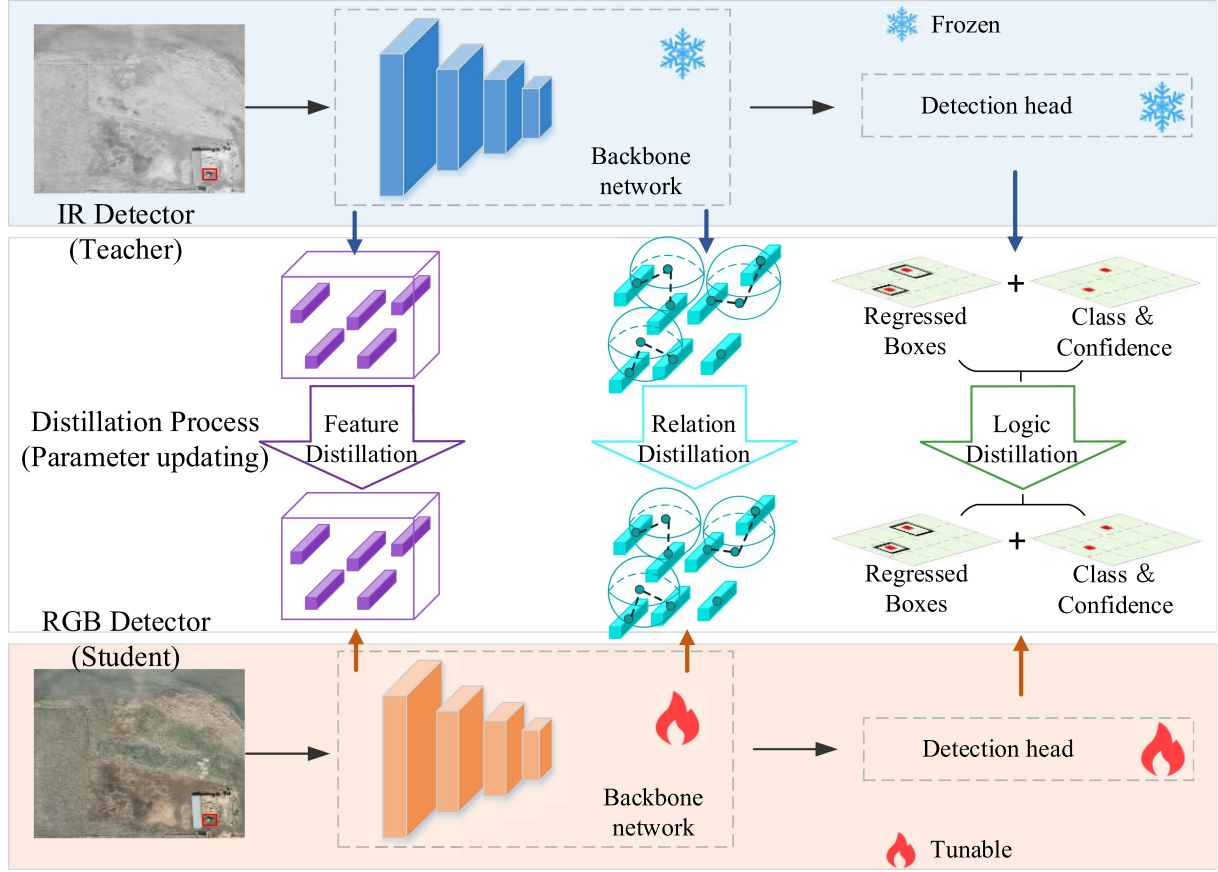


Fig. 3. General framework of the proposed CMDistill network. The training process consists of two stages. First, the single-modal model is trained and then the parameters of the teacher model (top) are frozen, and only the CMDistill parameters of the student model are updated. The inference process requires only RGB images as CMDistill inputs for object detection.

to achieve enhanced detection performance. First, to reduce the possible negative effects of large feature differences between different modalities, a PCC was calculated to achieve the feature knowledge transfer from intermediate layers in CKD. Second, to focus on the variability of the relationships between different modalities, a high-dimensional similarity loss was designed to transfer key relational knowledge. The soft-target knowledge predicted by the output layer of the teacher model plays a critical role in guiding the student model, given that the IoU between different modalities was calculated to transfer target knowledge in terms of localization and classification information. The proposed CMDistill framework is illustrated in Fig. 3 and includes three key components:

- 1) PCC feature distillation (PCCFD);
- 2) SLRD;
- 3) IBCLD.

A. Overall Framework

In CMDistill, the teacher and student models use the same target detector, which mainly consists of two key components: a feature extractor $f(\cdot)$ and detection head $h(\cdot)$. The detector extracts the feature $F = f(I_{\text{image}})$ using the feature extractor and then obtains the prediction $P = h(F)$ using the detection head.

The proposed CKD network consists of three modules: PCCFD, SLRD, and IBCLD, as shown in Fig. 3. Specifically, RGB and IR images are provided as inputs to the CMDistill teacher and student models, respectively. First, the backbone network processes the input image to extract representative features of the target, and then extracts the multiscale features of different modalities for knowledge transfer through a feature pyramid network (FPN). This article proposes a PCC to constrain the negative knowledge transferred from teacher to student during CKD and to reduce the influence of the student on the imitation of the teacher, which arises from intermodal differences (see Section III-B). We designed affinity matrices for high-level semantic features to compute the key information between the features of the teacher and student models to model the relational cues between features accurately (see Section III-C). Specifically, we designed an adaptive layer to map student features in the intermediate feature layer to reduce the difficulty of cross-modal feature imitation learning. Finally, for the features of dense and small targets in AAV aerial images, we designed an IoU-based logical distillation loss to pass the predicted bounding boxes and localization information from the output layer of the teacher model to the student, which improves the distillation effect of the student by combining multiple KD methods (see Section III-D). The updating of the student network model parameters

is supervised based on the detection and distillation losses (see Section III-E).

B. PCC Feature Distillation

Distillation using multiscale feature maps with rich spatial information to improve student models is a nonnegligible strategy for feature-layer distillation [15]. For a single-modal detector, it is reasonable to force students to imitate the features of the teacher, and multiscale feature imitation can lead to more desirable distillation results. However, in CKD, there are large differences between different modalities and significant noise, thus rendering it difficult to imitate feature maps directly to achieve an ideal detection effect. Therefore, we used a PCC for feature mimicry, with a focus on the linear correlations between features.

Notably, PCCFD imitates feature map loss at different scales and extracts knowledge representations from different detector modalities. However, several FPN feature maps may adversely affect the student model during the training phase. Therefore, the student is constrained to focus on critical information during feature map imitation and relax constraints on the distribution of different modal features. Specifically, we first process the feature maps by applying a normalization operation such that their mean and variance are zero, thus eliminating the negative effects of differences in the magnitudes of different modalities between feature maps at different scales. The mean-squared-error loss between the normalized features is then calculated. In particular, the PCC between the two original feature vectors is calculated as the feature mimicry loss. For a feature map of size $h \times w$, $s, t \in \mathbb{R}^m$ denotes the FPN output and we obtain the normalized values $\hat{t}_{1,\dots,m}$ and $\hat{s}_{1,\dots,m}$ for teachers and students, respectively. The PCCFD loss is calculated as follows:

$$\mathcal{L}_{\text{fea}} = r(s, t) = \frac{\sum_{i=1}^m (s_i - \mu_s)(t_i - \mu_t)}{\sqrt{\sum_{i=1}^m (s_i - \mu_s)^2} \sqrt{\sum_{i=1}^m (t_i - \mu_t)^2}} \quad (1)$$

where (s_i, t_i) are the m data points in the feature map $s, t \in \mathbb{R}^m$. The PCC is essentially a normalized measure of covariance with values in the range of $[-1, 1]$. Moreover, PCCFD lessens the restriction on the size of the features, with a focus on differences in the linear correlations between teacher–student features. Therefore, the PCC is suitable for cross-modal detector KD, thus bridging the gap between the different modalities of teachers and students to a certain extent.

C. Semantic-Level Relation Distillation

In addition to using PCCFD to address the direct distillation of features, the relationships between high-level semantic features contain information. The objective is to improve the distillation effect further by leveraging the relationships between features. The semantic features of different modalities are generally significantly different; thus, teacher and student features cannot be identical. The relative relationships obtained by directly computing high-level semantic features cannot reduce the gap between modalities. Therefore, to model the relational cues between features accurately, inspired by Hou et al. [47], we

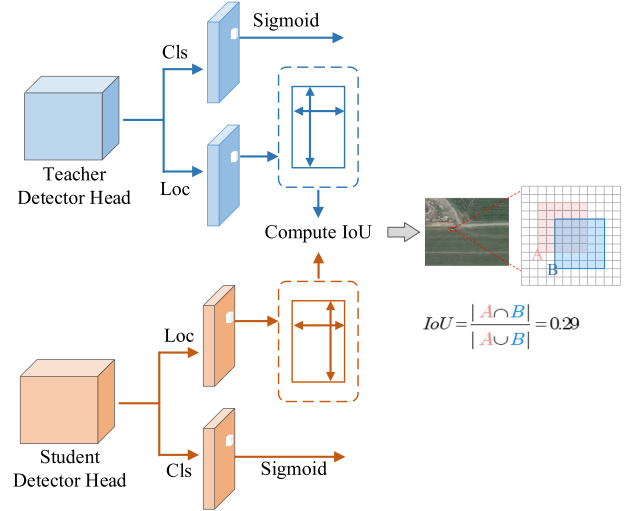


Fig. 4. Schematic of logic loss calculation based on the IoU.

construct affinity matrices of high-level semantic features to establish the relationships between them. Under the guidance of the high-level semantic features of the teacher, the feature similarities of different modalities can be improved. Specifically, we encode the similarity of each feature vector pair to compute the affinity matrix scores of the teacher and student models as follows:

$$A_{i,j} = \frac{f_i^T f_j}{\|f_i\|_2 \cdot \|f_j\|_2} \quad (2)$$

where f_i and f_j denote the i th and j th feature vectors, respectively. Subsequently, we use $L1$ loss to supervise students in learning the feature relation information of high-level semantic features from the teacher model as

$$\mathcal{L}_{\text{rela}} = \frac{1}{K \times K} \sum_{i=1}^K \sum_{j=1}^K \|A_{i,j}^t - A_{i,j}^s\|_1 \quad (3)$$

where K denotes the number of feature vectors. To reduce computational complexity, we compute only the affinity graph of the deepest semantic features of the model.

D. IoU-Based Binary Classification Logic Distillation

Logic-based KD is simple to use and effective at improving results because it guides the student model through the output layer of the teacher model and can generally be combined with other KD approaches. Therefore, we can further improve the distillation effect of the proposed model using logical information. In aerial images, most targets are small, and KD target detection is subject to a significant problem of imbalance between foreground and background. In addition, for the CKD target detection task, using the prediction results generated by the teacher model as soft labels to guide the student yielded a superior performance than using the ground truth to guide the student. This is because the structure between the teacher and student models is the same; thus, the student is more capable of mimicking the single-modal learning process of the teacher, as shown in Fig. 4.

To transfer the bounding box localization knowledge predicted by the teacher model to the student model without complex bounding box transformations, we use the bounding box IoU as a distillation target. Specifically, for a given input image $\{I_{\text{RGB}}, I_{\text{IR}}\}$, the teacher and student model predictions yield the predicted boxes $\text{Prob}_{\text{bbox}}^t$ and $\text{Prob}_{\text{bbox}}^s$ for the i th target. The localization information of $\text{Prob}_{\text{bbox}}^t$ and $\text{Prob}_{\text{bbox}}^s$ is represented by 4-D vectors containing information about the coordinates of the bounding box [i.e., $\text{Prob}_{\text{bbox}}^t = (x_1^t, y_1^t, x_2^t, y_2^t)$ and $\text{Prob}_{\text{bbox}}^s = (x_1^s, y_1^s, x_2^s, y_2^s)$]. The IoU between the two vectors is then calculated and the IoU distillation loss is defined as follows:

$$\mathcal{L}_{\text{IoU}} = \frac{\text{Prob}_{\text{bbox}}^t \cap \text{Prob}_{\text{bbox}}^s}{\text{Prob}_{\text{bbox}}^t \cup \text{Prob}_{\text{bbox}}^s} \quad (4)$$

where $\text{Prob}_{\text{bbox}}^t$ and $\text{Prob}_{\text{bbox}}^s$ denote the predicted rectangular boxes obtained via reasoning regarding the target by the teacher and student models, respectively. Furthermore, to solve the problem of a severe sample imbalance between a target and background in an aerial image, we modeled the multiclassification problem as multiple binary classifications [48] to allow for the student to inherit the predictive ability of the teacher. Specifically, the teacher and student models first calculate $p^{t'} = \text{Prob}(l^t)$ and $p^{s'} = \text{Prob}(l^s)$ to obtain the binary classification scores $p^{t'}$ and $p^{s'}$, respectively, where l^t and l^s denote the logits of the teacher and student models, respectively. The classification distillation loss is given by the following equation:

$$\begin{aligned} \mathcal{L}_{\text{BCE}} &= (p_{i,j}^{s'}, p_{i,j}^{t'}) \\ &= -((1 - p_{i,j}^{t'}) \cdot \log(1 - p_{i,j}^{s'}) + p_{i,j}^{t'} \cdot \log(p_{i,j}^{s'})) \end{aligned} \quad (5)$$

$$\mathcal{L}_{\text{cls}} = \sum_{i=1}^n \sum_{j=1}^K \mathcal{L}_{\text{BCE}}(p_{i,j}^{s'}, p_{i,j}^{t'}) \quad (6)$$

where $\mathcal{L}_{\text{cls}}(\cdot)$ is the categorical distillation loss, $\mathcal{L}_{\text{BCE}}(\cdot)$ is the binary cross-entropy loss, and $p_{i,j}^{t'}$ and $p_{i,j}^{s'}$ are the i th and j th categories of $p^{t'}$ and $p^{s'}$, respectively. The logical distillation loss $\mathcal{L}_{\text{log}}$ consists of the prediction frame distillation loss $\mathcal{L}_{\text{IoU}}$ and categorical distillation loss $\mathcal{L}_{\text{cls}}$, as follows:

$$\mathcal{L}_{\text{log}} = \mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{cls}}. \quad (7)$$

E. Total Loss

In the training phase, the network parameters of the student model are optimized by a combination of the detection loss $\mathcal{L}_{\text{det}}$ and distillation loss. The total loss function $\mathcal{L}_{\text{Total}}$ is defined as follows:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{det}} + \lambda_1 \cdot \mathcal{L}_{\text{fea}} + \lambda_2 \cdot \mathcal{L}_{\text{rela}} + \lambda_3 \cdot \mathcal{L}_{\text{log}} \quad (8)$$

where $\mathcal{L}_{\text{det}}$ denotes the detection loss of the student model, and $\mathcal{L}_{\text{fea}}$, $\mathcal{L}_{\text{rela}}$, and $\mathcal{L}_{\text{log}}$ denote the feature, relationship, and logic losses of KD, respectively. Moreover, λ_1 , λ_2 , and λ_3 are the hyperparameters that compensate for different losses. The parameter updating process for the student model is summarized in Algorithm 1. It should be noted that, during the optimization of the student model, the weights of the teacher model are frozen, and no further updating of its parameters is required.

Algorithm 1: Student (RGB) Model Parameter Updating Process.

Input:

$\{\text{Image}\}$: RGB, IR images

θ_{IR} : trained IR target detector (parameter update not required)

θ_{RGB} : RGB target detector with initialized parameters

n : Total number of rounds of iterative optimization of parameters; η : learning rate; y : label

Output: optimized θ_{RGB}

for $i = 0$ to n do

input image pair $(X_{\text{IR}}^i, X_{\text{RGB}}^i) \in \{\text{Image}\}^i$

use teacher model θ_{IR} to obtain the intermediate-layer feature $\text{fea}_{\text{IR}}^{\text{IR}}$ of input image X_{IR}^i

use student model θ_{RGB} to obtain the intermediate-layer feature $\text{fea}_{\text{RGB}}^{\text{RGB}}$ of input image X_{RGB}^i and the predicted output $\hat{y}$

calculate the loss of the student model using (8) and

update its parameters θ_{RGB} as

$$\theta_{\text{RGB}} \leftarrow \theta_{\text{RGB}} - \eta \frac{\partial \mathcal{L}}{\partial \theta_{\text{RGB}}}$$

end for

For testing, we remove all the parameters of the IR detector, PCCFD, SLRD, and IBCLD. We can achieve real-time target detection in AAV images by inputting only RGB images and using RGB detectors. In particular, we used only RGB single-modal detectors in the AAV. Therefore, the AAV does not require IR sensors, which reduces the cost of the AAV and lowers the power consumption and computational requirements of the AAV edge-computing platform. In this study, extensive experiments were conducted to demonstrate the effectiveness of the proposed scheme.

IV. EXPERIMENTS

This section describes the experimental details and results. Section IV-A describes the experimental implementation details, including the dataset and evaluation metrics used. Section IV-B presents a quantitative and qualitative evaluation of the proposed method in comparison with other SOTA methods. Section IV-C presents the extensive ablation studies conducted to verify the validity of the proposed framework.

A. Implementation Details

We evaluated the performance of CMDistill on the VEDAI benchmark dataset and compared it with other representative SOTA methods. These representative methods are mainly classified into RGB detection methods and KD-based detection methods. The SOTA RGB target detection algorithms were the fully convolutional one stage (FCOS) [49], task-aligned one-stage object detection (TOOD) [50], Gaussian-receptive-field-based label assignment (RFLA) [51], YOLOv5 [52], YOLOv8 [52], and real-time detection transformer (RT-DETR) [53] models. The SOTA KD target detection algorithms were channelwise KD (CWD) [17], KD via PCC (PKD) [15], bridging cross-task KD (BCKD) [48], and masked generative distillation (MGD)

[16]. It should be noted that all compared methods used default hyperparameter settings, and the other KD methods were implemented utilizing the proposed backbone.

1) *Dataset*: To validate the performance of the proposed CMDistill, we evaluated it on the VEDAI dataset [54]. The VEDAI dataset consists of images collected at the same altitude as the automated georeferencing center in Utah, where each pixel in each image represents a physical space of approximately $12.5 \text{ cm} \times 12.5 \text{ cm}$. The scenes include a variety of complex backgrounds, such as motorways and urban areas. The dataset contains 1246 aligned RGB-IR image pairs and the detection targets include 11 different types of vehicles (e.g., cars, pickups, campers, and trucks). We divided the training and testing sets at an approximate ratio of 8:2. In addition, the DroneVehicle dataset [36], which includes aerial images of targets, such as cars, trucks, vans, buses, and lorries, allows for further evaluation of the detection performance of different methods. This dataset includes both RGB-T modalities and includes challenging scenarios, such as low light and occlusion.

2) *Evaluation Metrics*: We evaluated the algorithms using common target detection evaluation metrics. These metrics include mean average precision (mAP), precision, recall, and precision–recall (PR) curves. In particular, mAP evaluates the mean detection accuracy of a model and is defined as follows:

$$\text{AP} = \int_0^1 \text{Precision} d(\text{Recall}) \quad (9)$$

$$\text{mAP} = \frac{\sum_{i=1}^K \text{AP}_i}{K} \quad (10)$$

where K is the number of categories in the target detection task. In addition, to characterize the dynamic relationship between precision and recall, we plotted PR curves for the different algorithms. These metrics are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (11)$$

where TP, FP, TN, and FN are the true-positive, false-positive, true-negative, and false-negative rates, respectively.

3) *Experimental Details*: We used YOLOv5 [52], which is a one-stage target detector with an FPN layer, as the backbone network for the teacher (IR detector) and student models (RGB detector). For the training of the proposed model, the image is resized to 640×640 pixels and data enhancement is then applied, which mainly consists of random rotation, random crop, and color dithering. The stochastic gradient descent method was used to optimize the network parameters, with the initial learning rate and momentum set to 0.01 and 0.937, respectively. The batch size was set to 64 and the weight decay was set to $5e-4$. The learning rate was reduced using a cosine decay strategy. In the training phase, we used images from two different modalities (RGB and IR) to train the student and teacher models. In the inference phase, only a single modality is used for testing. All methods were implemented on the PyTorch deep learning framework and tested on a small deep learning workstation with an Intel i9-13900K at 5.80 GHz CPU and GeForce RTX 4090 GPU.

B. Comparative Experiment

To demonstrate the superiority of CMDistill, we performed extensive comparisons with other SOTA methods on the VEDAI and DroneVehicle datasets, including quantitative and qualitative visual evaluations.

1) *Quantitative Results*: Table I presents the mAP comparisons of the 11 SOTA methods on the VEDAI dataset. **Bold red** indicates the largest value in each column and **bold blue** indicates the second largest value. It can be seen that CMDistill achieved the best average detection accuracy. Compared with the single-modal target detection algorithms, the performance improvement was significant, given that the mAP value of CMDistill was 26% higher than that of FCOS. Compared with TOOD, which had the second-optimal performance, the mAP value was improved by 0.4% and the accuracy of the other categories improved to varying degrees. It should be noted that the baseline for the teacher model was YOLOv5s, and the performance improvement achieved by CMDistill was significant, with a 3.8% improvement in the mAP when compared with the original single-modal teacher model. The different degrees of improvement in detection accuracy for different classes of targets fully demonstrate the effectiveness of CKD, which can improve the performance of a target detection framework without increasing model complexity. In addition, the proposed method was demonstrated as significantly superior to the most popular target detectors, namely, YOLOv8 and RT-DETR. In particular, YOLOv8 is a SOTA algorithm for industrial applications, and the proposed framework yields an mAP value of 3.1% higher than that of YOLOv8. The detection accuracy was higher than that of YOLOv8 for all target types, except for tractors and vans. Moreover, RT-DETR is a detection algorithm based on the transformer architecture, which demonstrated a higher inference speed. However, it is slightly inferior to the proposed method in terms of detection accuracy. This is because most of the targets in the VEDAI dataset are significantly small and the features obtained by relying only on RGB images are limited. Moreover, CMDistill learns the IR feature knowledge conveyed by the teacher model through CKD during training and exploits information from two different modalities to enable the enhanced characterization of target features, thus resulting in improved target detection performance.

The proposed CMDistill method demonstrated a superior detection performance when compared with other SOTA KD target detection algorithms. Moreover, when compared with CMDistill, the mAP values of the CWD, PKD, and MGD methods were 1.3%, 4.5%, and 3.3% lower, respectively, given that these methods only consider knowledge transfer in the intermediate feature layers of their networks. For the CKD task, the predictive knowledge in the output layer contained the target category and localization information, which are crucial for improving target detection performance. The mAP of BCKD was 1% lower than that of CMDistill because BCKD only considers knowledge transfer in the output layer, and the cross-modal feature knowledge in intermediate layers is relevant for improving model performance. In conclusion, the above-mentioned experimental results reveal that CMDistill demonstrated a high detection

TABLE I
COMPARISON WITH SOTA METHODS ON THE VEDAI DATASET

Method	Modality	Car	Pickup	Camper	Truck	Other	Tractor	Boat	Van	mAP	Params/M	FPS
FCOS	RGB	0.661	0.628	0.647	0.358	0.338	0.400	0.548	0.257	0.480	31.85	53.6
TOOD	RGB	0.919	0.842	0.862	0.679	0.570	0.839	0.593	0.584	0.736	31.81	46.8
RFLA	RGB	0.897	0.792	0.742	0.446	0.516	0.692	0.642	0.294	0.627	31.85	55.5
YOLOv5s	RGB	0.885	0.839	0.721	0.654	0.601	0.781	0.703	0.433	0.702	9.14	256.4
YOLOv8s	RGB	0.862	0.799	0.727	0.587	0.543	0.877	0.673	0.608	0.709	11.16	550.1
RT-DETR	RGB	0.89	0.833	0.785	0.65	0.597	0.653	0.766	0.636	0.726	19.97	67.8
CWD	$C_{T_{IR} \rightarrow S_{RGB}}$	0.903	0.866	0.678	0.684	0.649	0.758	0.692	0.589	0.727	7.03	277.8
PKD	$C_{T_{IR} \rightarrow S_{RGB}}$	0.891	0.857	0.667	0.648	0.586	0.799	0.672	0.443	0.695	7.03	277.8
BCKD	$C_{T_{IR} \rightarrow S_{RGB}}$	0.889	0.852	0.71	0.644	0.682	0.829	0.7	0.538	0.73	7.03	277.8
MGD	$C_{T_{IR} \rightarrow S_{RGB}}$	0.864	0.831	0.694	0.67	0.636	0.855	0.72	0.389	0.707	7.03	277.8
Ours	$C_{T_{IR} \rightarrow S_{RGB}}$	0.907	0.86	0.708	0.701	0.65	0.87	0.68	0.543	0.740	7.03	277.8

$C_{T_{IR} \rightarrow S_{RGB}}$ denotes the CKD (teachers modeled as IR and students modeled as RGB) method.

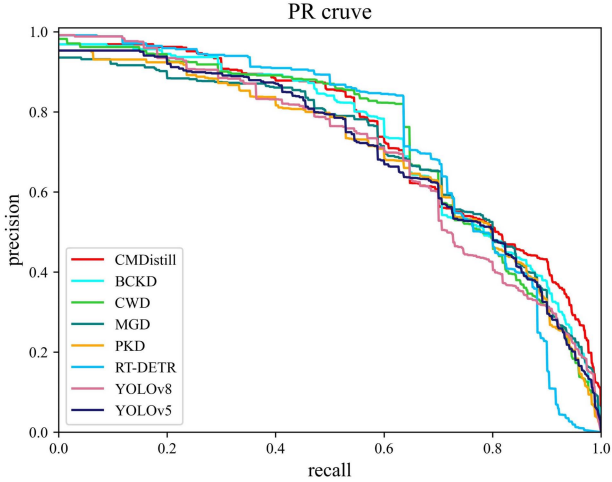


Fig. 5. Comparison of the PR curves of CMDistill and other methods. Our architecture provides the best precision and recall.

performance by considering both intermediate-layer knowledge and output-layer knowledge transfer. This verifies that, for the CKD target detection task, multiple knowledge transfers should be considered. First, it is necessary to focus on the simple and effective transfer of prediction knowledge from the output layer. In addition, the transfer of feature knowledge in the intermediate layer should be considered. It is essential to combine and transfer such knowledge effectively.

Fig. 5 presents a comparison of the PR curves of CMDistill and the other methods on the VEDAI dataset. It can be seen that CMDistill demonstrated SOTA-level precision and recall on the VEDAI dataset, thus outperforming other methods. Jitter in the PR curve of RT-DETR is evident, which indicates that this model could not adapt well to the varied and complex backgrounds. The VEDAI dataset contains numerous suspected targets; thus, it was challenging for the detection algorithms to handle. The

TABLE II
COMPARISON WITH SOTA METHODS ON THE DRONEVEHICLE DATASET

Method	Modality	Car	Freight-Car	Truck	Bus	Van	mAP
Faster R-CNN	RGB	79.0	37.2	49.0	77.0	37.0	55.9
RoITransformer	RGB	61.6	42.3	55.1	85.5	44.8	61.6
YOLOv5s	IR	90.0	60.8	59.5	89.5	53.8	70.7
RoITransformer	IR	90.1	58.9	60.4	89.7	52.2	70.3
UA-CMDet	RGB+IR	88.6	57.0	73.1	88.5	54.1	70.0
MBNet	RGB+IR	90.1	62.4	64.4	88.8	53.6	71.9
AFFCM	RGB+IR	90.2	64.9	73.4	89.9	64.9	76.6
TSFADet	RGB+IR	89.9	63.7	67.9	89.8	54.0	73.1
C ² Former	RGB+IR	90.2	64.4	68.3	89.8	58.5	74.2
M2FP	RGB+IR	95.7	64.7	76.2	92.1	64.7	78.7
Ours	$C_{T_{IR} \rightarrow S_{RGB}}$	94.4	56.7	75.1	94.4	50.8	74.3

$C_{T_{IR} \rightarrow S_{RGB}}$ denotes the CKD (teachers modeled as IR and students modeled as RGB) method.

unsatisfactory performance of the other CKD target detection methods is related to their limited knowledge transfer features. The results reveal that CMDistill is robust to various complex scenarios and provides improved detection performance, which allows for the realization of potential applications.

Furthermore, to demonstrate the sophistication and generalizability of the proposed methods, we compared them with the SOTA methods on the DroneVehicle dataset. The methods compared mainly included the unimodal target detection methods: Faster R-CNN [23], RoITransformer [55], and YOLOv5s [52], and the RGB-T-based multimodal fusion target detection methods: UA-CMDet [36], MBNet [56], AFFCM [57], TSFADet [7], C²Former [5], and M2FP [58]. The experimental results are shown in Table II. As can be seen from the table,

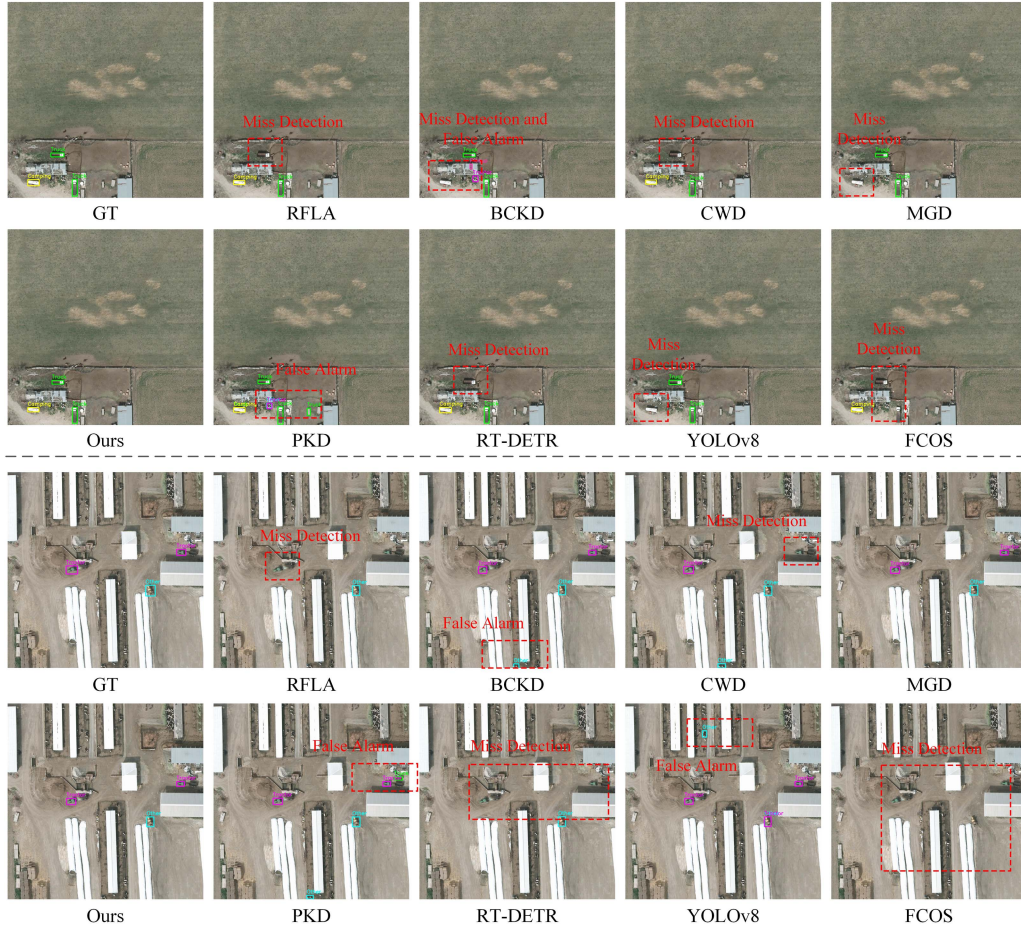


Fig. 6. Visualization of the detection results obtained by different algorithms on the VEDAI dataset, with different target types marked in different colors. Images are zoomed for better viewing.

CMDistill demonstrated a superior performance to all other unimodal methods, given that the proposed method effectively improves the detection performance of the unimodal model by transferring the IR feature knowledge through the teacher model. The proposed method demonstrated a superior detection performance to those of the RGB-T-based methods, such as UA-CMDet, MBNet, and TSFADet. Although the proposed method is slightly inferior to AFFCM and M2FP, the latter requires RGB and IR images as input and more complex models. The results reveal that CMDistill demonstrates a high potential for the effective characterization of IR image features through KD.

2) *Visualization of Results:* To illustrate the advantages of CMDistill visually, we compared the experimental results of CMDistill with those of other SOTA methods, as shown in Fig. 6. It can be seen that, in the challenging scenarios with grass and building backgrounds, there were multiple suspected target components, and it was difficult to distinguish between the actual and suspected targets.

All methods, with the exception of CMDistill, exhibited varying degrees of missed detections and false alarms. In the first group of images, BCKD and PKD exhibited the most false alarms, whereas FCOS exhibited the most missed detections.

These methods lack an effective feature identification mechanism to accurately differentiate suspected targets, thus resulting in poor detection performance when handling a complex building background. In the second group of images, FCOS did not highlight targets, RT-DETR and YOLOv8 exhibited false alarms and missed detections, and the other CKD methods demonstrated equally poor performances. These results demonstrate that the existing methods are less robust under challenging conditions and cannot adapt well to changes in complex backgrounds. In contrast, CMDistill learns the representative knowledge conveyed by different modalities through the feature layer and output layer, thus enabling it to characterize the representative features of targets through CKD, which results in excellent detection performances in complex scenes.

C. Ablation Experiments

To validate the potential benefits of the key components of the proposed CMDistill method, we performed a detailed ablation study on the VEDAI dataset.

1) *Effectiveness of the Logical-Layer Distillation Design:* This section describes the ablation experiments conducted on

TABLE III
ABLATION STUDY ON DIFFERENT DISTILLATION LOSSES ON THE VEDAI DATASET

Setting	$\mathcal{L}_{\text{fea}}$	$\mathcal{L}_{\text{rela}}$	$\mathcal{L}_{\text{log}}$	Precision	Recall	mIoU	mIoU0.5-0.95
1				0.714	0.592	0.702	0.417
2	✓			0.732	0.622	0.709	0.418
3		✓		0.757	0.598	0.713	0.404
4			✓	0.724	0.602	0.728	0.413
5	✓	✓		0.700	0.671	0.719	0.433
6	✓		✓	0.666	0.640	0.729	0.452
7		✓	✓	0.700	0.640	0.730	0.439
8	✓	✓	✓	0.707	0.655	0.740	0.456

“T” and “S” represent the teacher detector and student detector, respectively. The highest value in each column is shown in bold.

TABLE IV
ABLATION STUDY ON THE SELECTION OF DIFFERENT FEATURE LAYERS FOR KD

Setting	C5	P4	P3	P2	Precision	Recall	mIoU	mIoU0.5-0.95
1	✓				0.705	0.644	0.738	0.453
2		✓		✓	0.702	0.652	0.734	0.449
3	✓	✓	✓		0.695	0.665	0.734	0.447
4	✓	✓	✓	✓	0.713	0.639	0.735	0.446
5	✓			✓	0.707	0.655	0.740	0.456

The highest value in each column is shown in bold.

the IBCLD to investigate the effect of logical-layer distillation on the distillation effectiveness of CMDistill. As shown in Table III, the use of logic distillation improves the target detection performance of the student model. Compared with the model without KD, the target detection performance was significantly improved when using the IBCLD, with a 3.8% increase in the mAP value. In addition, the other settings in Table III demonstrate the ability of logical components to serve as effective complements to the losses of feature distillation and relational distillation. This is because logic distillation transfers knowledge through the output layers of the teacher and student models, which is not dependent on the intermediate-layer features, and allows for the student model to locate target positions in images. Overall, the experimental results demonstrate that the IBCLD designed in this study plays a critical role in the KD of CMDistill and facilitates knowledge transfer from the teacher model in different modalities.

2) Effectiveness of the Feature-Layer Distillation Design:

This part describes the ablation experiments conducted on the PCCFD to investigate the effects of feature-layer distillation on the KD performance of CMDistill. As shown in Setting 2 in Table III, the use of feature-layer distillation can help improve

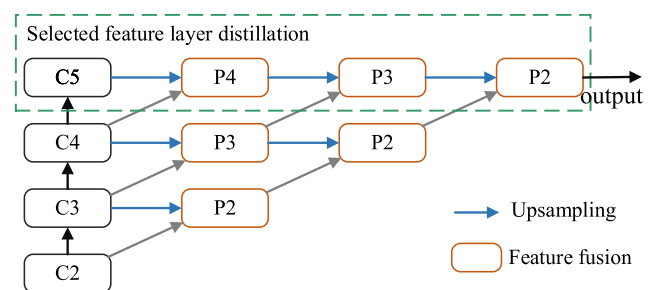


Fig. 7. Schematic representation of feature-layer selective distillation. The green rectangular box represents the selective distillation of feature maps. Blue arrows and orange rectangular boxes indicate the upsampling and feature map fusion, respectively.

the target detection capacity of the student model. Although the distillation effect of the PCCFD under Setting 2 was not significant, it could not be neglected because it complemented the other distillation losses under Settings 5, 6, and 8, thus effectively improving the poor performances when using other distillation losses. The results in Table III are reasonable when considering that the sole dependence on feature layers to perform CKD is

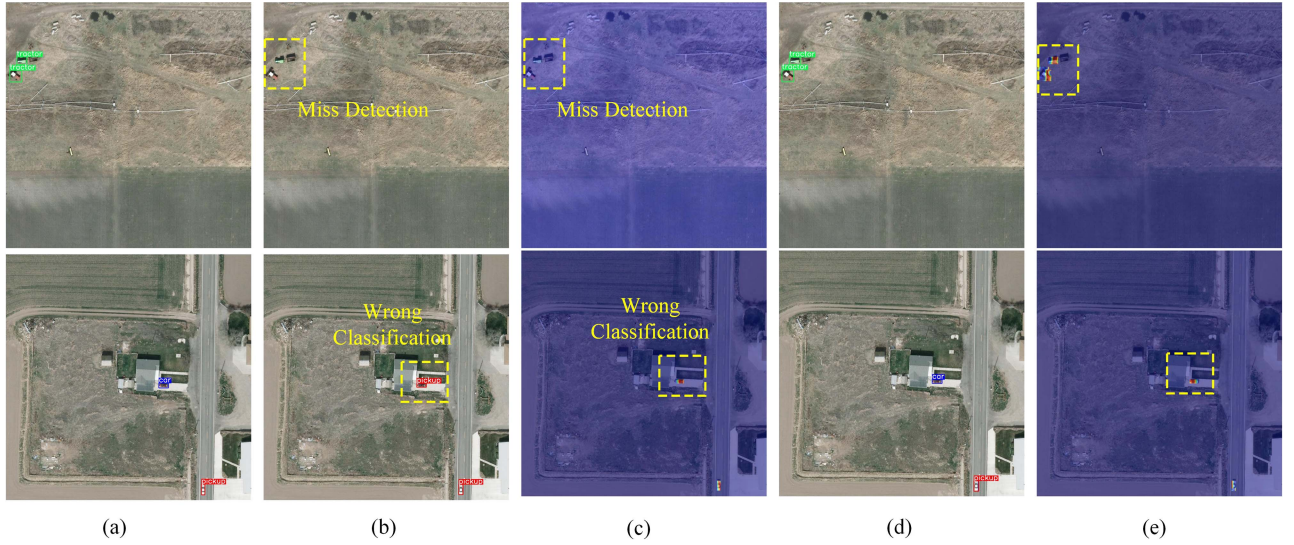


Fig. 8. Visualization of detection results. The red and green boxes represent the predicted and true bounding boxes, respectively. (a) Ground truth. (b) and (d) Results of the student detector without CMDistill and with CMDistill, respectively. (c) and (e) Results of heat map visualization for the student detector without CMDistill and with CMDistill, respectively. It can be seen that CMDistill helps the student detector to produce more accurate results and reduces false alarms. (a) Ground truth. (b) Student w/o CMDistill. (c) Student w/o CMDistill heatmap visual. (d) Student with CMDistill. (e) Student with CMDistill heatmap visual.

highly challenging, owing to large modal differences between the teacher and student model training data. This confirms that the PCCFD designed in this study can be useful for CKD tasks. The feature-layer selective distillation is illustrated in Fig. 7.

In addition, we compared the distillation results of different feature layers to validate the selection of feature layers used in the experiments. The results in Table IV illustrate that fewer feature layers were more effective for feature-layer distillation in both the student and teacher models. As shown for Setting 4, using all feature layers yielded poor KD results. This can be attributed to large differences between the feature layers of different modalities. Moreover, using an excessively high number of feature layers may introduce noise, thus leading to performance degradation. We selected the deepest and shallowest feature layers for CKD to achieve enhanced distillation results, and using fewer feature distillation layers resulted in higher efficiency during training.

3) *Effectiveness of the SLRD Design*: This part details the ablation experiments conducted on SLRD to investigate the effects of the relational-layer distillation on the distillation effectiveness of CMDistill. As shown under Setting 3 in Table III, the use of logic distillation improved the target detection performance of the student model. Compared with the model without any KD, the target detection performance was improved with SLRD with a 1.1% increase in the mAP value. In addition, the experimental results for Settings 5, 7, and 8 in Table III demonstrate the capacity of relational losses to be combined with other distillation losses, which can effectively improve the target detection accuracy of the student model. This is because for the CKD target detection task, establishing relationships between the intermediate feature layers of students and teachers in different modalities can improve the characterization and transfer of knowledge, thereby improving the target detection performance of the student model.

To visualize the effectiveness of the proposed KD loss design, we visualized the results of target detection with and without KD loss training. The experimental results are shown in Fig. 8. Fig. 8(a) presents the ground-truth images, and Fig. 8(b) and (d) presents the results of the student detector without and with CMDistill, respectively. Fig. 8(c) and (e) presents the results of heat map visualization for the student detector without and with CMDistill, respectively. As shown in Fig. 8(c), the model without CMDistill was unable to locate the target in the first plot accurately. Therefore, there was a missed detection. As can be seen from the second image, although the target was localized, it could not be accurately classified. Therefore, the results, as shown in Fig. 8(b), are reasonable. Fig. 8(e) reveals that the proposed method can accurately locate and classify the target. These detection results strongly support the effectiveness of the CMDistill process, which can effectively improve the accuracy of the CKD target detection algorithm.

4) *Ablation Experiment on Adaptive-Layer Necessity*: Before calculating the distillation loss using the feature maps corresponding to the teacher model, we used an adaptive layer to map student features. This operation is necessary for cross-modal distillation and reduces the difficulty of cross-modal feature imitation learning, which improves the target detection accuracy. We verified the critical role of the adaptive layer in CMDistill by evaluating the model after removing this layer. As can be seen from Fig. 9, the removal of the adaptive layer resulted in a significant decrease in the detection performance. This is a result of the large gap between the different modalities used in CKD. The student features in the intermediate layers should be adjusted by the adaptive layer to match the teacher's features. Without an adaptive layer, the direct matching of student and teacher features degrades the performance. Therefore, it can be concluded that the adaptive layer effectively improves the detection performance of the student mode.

TABLE V
ABLATION STUDIES ON DIFFERENT MODAL TEACHER AND STUDENT MODELS

Modality (S)	Modality (T)	Precision	Recall	mAP	mAP _{0.5-0.95}
RGB	—	0.714	0.592	0.702	0.417
IR	—	0.661	0.580	0.656	0.401
RGB+IR	—	0.671	0.678	0.717	0.437
IR	RGB+IR	0.704	0.607	0.704(↑4.8%)	0.433(↑3.2%)
RGB	RGB+IR	0.667	0.678	0.736(↑3.4%)	0.444(↑2.7%)
IR	RGB	0.706	0.605	0.693(↑3.7%)	0.431(↑3 %)
RGB	IR	0.707	0.655	0.740(↑3.8%)	0.456(↑3.9%)

where “T” and “S” represent the teacher detector and student detector, respectively.

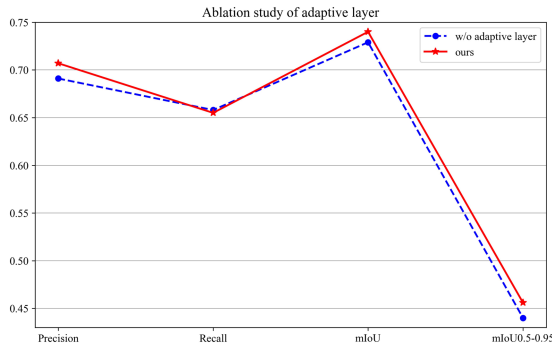


Fig. 9. Results of the ablation study on adaptive feature layers.

5) Ablation Experiments for Different Modal Distillations:

To demonstrate that the proposed CMDistill is not limited to KD from IR to RGB modalities, we verified that it can demonstrate generalization for different modal realizations of CKD. Specifically, we investigated the effects of different teacher models on KD. We selected RGB and RGB-T teacher models, and IR and RGB student models. As can be seen from the experimental results, as shown in Table V, when the teacher modality was IR, the mAP value of the RGB student model increased by 3.8%, and when the teacher modality was RGB, the mAP value of the IR student model increased by 3.7%. When the teacher modality was RGB + IR, the mAP values of the IR and RGB student models increased by 4.8% and 3.4%, respectively. These results demonstrate that the information on different modalities or multiple modalities embedded in the teacher model significantly improved the detection performance of single-modal student models. Therefore, it can be concluded that the CMDistill method provides adequate generalization for the CKD target detection task for different modalities. For a teacher model with different modalities, its capacity to transfer knowledge through the proposed CKD approach effectively improves the performance of the student model, thus serving as a basis for the improvement of single-modal target detection performance. In addition, we found that the use of CKD to improve single-modal target detection performance is a potential method that avoids the complexity of RGB-T model deployment and can achieve a superior tradeoff between

accuracy and speed for the deployment of AAV edge-computing platforms.

D. Discussion

The cross-modal KD framework is a preferable solution, owing to the following. First, actual scene data sources are diverse, and it is difficult to process multimodal data simultaneously, e.g., AAV payloads include visible, IR, and LiDAR cameras, among others. Processing these data simultaneously consumes a large amount of computational resources, which edge-computing platforms cannot handle. In addition, mounting multiple types of sensors significantly increases the AAV cost. Second, the teacher model obtained using a large amount of multimodal data training demonstrates a higher detection performance and generalization capacity. Knowledge transfer through a cross-modal KD framework not only reduces the demand for sensors and edge-computing resources but also effectively improves the detection performance of unimodal models.

With respect to the limitations of the study and scope of future research, the proposed cross-modal KD framework only considers the transfer of IR feature knowledge through the teacher model. The performance of multimodal feature fusion as a teacher model for knowledge transfer was not investigated, which may be advantageous with respect to missing modalities. Furthermore, despite the high detection performance achieved by CMDistill, there were several false positives, and there is still room for improvement. It should be noted that there are multiple modal representations for most of the data. Therefore, future research should involve the introduction of novel feature distillation strategies to provide more comprehensive multimodal features to the student model and the investigation of the ability of cross-modal frameworks to generalize to unseen modalities.

V. CONCLUSION

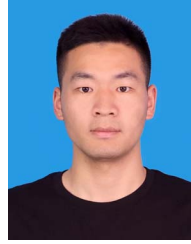
In this study, we developed the CMDistill method to improve the performance of a single-modal target detector by optimizing the tradeoff between accuracy and complexity. The proposed method is simple and readily deployable on AAV edge-computing platforms. In addition, CMDistill does not require the consideration of complex multimodal fusion designs and can achieve a superior detection performance to that of

a conventional single-modal detector by incorporating three types of distillation loss. In addition, the design of the CKD adaptive layer can filter unaligned background information and reduce the noise interference caused by modal differences. The results of numerous experiments demonstrated that the CMDistill framework can effectively improve the performance of student detectors. In particular, CMDistill achieved an mAP of 74% in the RGB-only target detection task, which was 3.8% higher than the original single-modal teacher model. Based on the merits of CKD, we will further investigate the use of modal correlation for cross-modal target detector performance enhancement and a novel target detection framework for CKD that can be deployed on AAVs.

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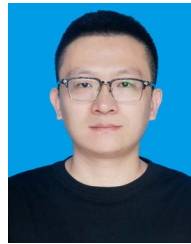
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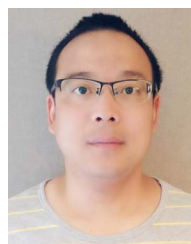
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